

SCILLA PASSIVE

Industrial Evidence Brief - Public Evidence Release 1.0.0

Hypothesis. Use mechanically scanning merchant radars as non-cooperative, moving illumination opportunities for a track that already exists. The product hypothesis is a passive *second-observation* layer, not a replacement radar.

Evidence status. Simulation/public research only. No measured RF performance.

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Policy	Median final error	95% CI
No passive	351.5 m	327.2–396.4 m
Shortest pulse	19.5 m	17.4–22.2 m
Metrology-conditioned EIG	19.2 m	16.9–21.8 m

Hard result. Passive updates beat propagation-only in 94.4% of 54 physics stress cells. Three failure cells are preserved.

Hard negative result. The EIG policy does not robustly beat shortest-pulse selection; it wins 49.7% of paired nominal worlds. **Optimizer moat not demonstrated.**

What remains unproven. Sea-clutter suppression, direct-path dynamic range, emitter association, real bistatic RCS, field geolocation, customer economics and operational utility.

Why an industrial partner matters. A radar/UAS/surveillance organization already possessing RF labs or archived mission data can close the dominant evidence gap at low marginal cost through independent replay or partner-sponsored receive-only validation.

Diligence object. Frozen equations + simulator + 300-world paired benchmark + 54-cell physics grid + 27-cell integrity grid + preserved failure history + source ledger + reproduction protocol.

Next external gate. Independent reproduction / customer replay / partner-sponsored RF validation.

No traction, acquisition interest, patentability or field performance is asserted.